

Mohammad Mahmudul Hasan, PhD

Associate Professor

Department of Electrical and Electronic Engineering



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EDUCATION

-  **Norwegian University of Science and Technology, Norway** (05.2022 – 06.2025)
Doctor of Philosophy
Information and Communication Technology
-  **KIIT University, India** (07.2008 – 06.2010)
Master of Technology
Communication Systems Engineering
-  **KIIT University, India** (07.2004 – 06.2008)
Bachelor of Technology
Electronics and Telecommunication Engineering

WORK EXPERIENCE

-  **University of Information Technology and Science, Bangladesh** (01.2011 – Present)
 - ✓ *Associate Professor*, Electrical and Electronic Engineering 07.2025 – Present
 - ✓ *Assistant Professor*, Electrical and Electronic Engineering 01.2011 – 06.2025
-  **Norwegian University of Science and Technology, Norway** (05.2022 – 06.2025)
 - ✓ *Research Fellow*, Faculty of Engineering
-  **Brno University of Technology, Czech Republic** (08.2024 – 01.2025)
 - ✓ *Visiting Research Fellow*, Department of Automation
-  **KIIT University, India** (02.2009 – 12.2010)
 - ✓ *Assistant Professor*, Electronics and Telecommunication Engineering 07.2010 – 12.2010
 - ✓ *Teaching Assistant*, Electronics and Telecommunication Engineering 02.2009 – 06.2010

SELECTED PUBLICATIONS

-  **Mohammad Mahmudul Hasan**, “MWFedNet: Federated Learning for Heterogeneous Microwave Sensing Networks”, **IEEE Transactions on Neural Networks and Learning Systems**, under review, 2026 (2026)
-  Inaamullah Khan, **Mohammad Mahmudul Hasan**, Michael Cheffena, “Transmitter-Assisted Joint Data-Aided Channel Estimation and PAPR Reduction Scheme in Wireless Fading Channels”, **Scientific Reports**, vol. 16, no. 1, 8015, 2026 (2026)
-  **Mohammad Mahmudul Hasan**, Onur Alev, Pavel Skrabanek, Gabriela Soukupová, Fatima Hassouna, and Michael Cheffena, “Microwave MIMO E-Nose for Wireless Communication and Selective Detection of VOC Mixtures with Concentration Estimation”, **ACS Sensors**, vol. 10, no. 9, pp. 6446–6463, 2025 (2025)
-  Inaamullah Khan, **Mohammad Mahmudul Hasan**, Michael Cheffena, “A Novel Low-Complexity Peak-Power-Assisted Data-Aided Channel Estimation Scheme for MIMO-OFDM Wireless Systems”, **IEEE Open Journal of Signal Processing**, vol. 6, pp. 992 – 1003, 2025 (2025)
-  Onur Alev, **Mohammad Mahmudul Hasan**, Emel Tuğba Ertuğrul, Selçuk Birdoğan, Okan Özdemir, Eda Goldenberg, Michael Cheffena, “Hydrothermally Synthesized Molybdenum disulfide Nanoflakes: Structural, Electrical, and Antenna-based Gas Sensing Characteristics”, **Sensors & Actuators: A. Physical**, vol. 393, 116756, 2025 (2025)
-  **Mohammad Mahmudul Hasan**, Onur Alev, Pavel Skrabanek, and Michael Cheffena, “Molecularly Imprinted Polymer-Based Electronic Nose for Ultrasensitive, Selective Detection and Concentration Estimation of VOC Mixtures”, **IEEE Sensors Journal**, vol. 25, no. 10, 2025 (2025)
-  **Mohammad Mahmudul Hasan**, Todd Cowen, Onur Alev and Michael Cheffena, “MIMO Microwave Sensor for Selective and Simultaneous Detection of Methanol and Ethanol Gases at Room Temperature,” **IEEE Transactions on Instrumentation & Measurement**, vol. 74, 9511613, 2025 (2025)

- 📄 **Mohammad Mahmudul Hasan**, Onur Alev and Michael Cheffena, “Dual-Functional Antenna Sensor for Highly Sensitive and Selective Detection of Isopropanol Gas Using Optimized Molecularly Imprinted Polymers,” *ACS Sensors*, vol. 10, no. 3, pp. 2147–2161, 2025 (2025)
- 📄 **Mohammad Mahmudul Hasan**, Onur Alev, Eda Goldenberg and Michael Cheffena, “MoS₂/MoO_x Nanoflake-Based Dual-Functional Antenna Sensors for Highly Sensitive and Selective Detection of Volatile Organic Compounds,” *ACS Applied Nano Materials*, vol. 7, no. 21, pp. 25065–25077, 2024 (2024)
- 📄 **Mohammad Mahmudul Hasan**, Todd Cowen and Michael Cheffena, “A Novel Molecularly Imprinted Polymer-Based Carbon Nanotube-Coated Microwave Sensor for Selective Detection of Methanol Gas,” *IEEE Sensors Letters*, vol. 8, no. 5, 6004904, 2024 (2024)
- 📄 **Mohammad Mahmudul Hasan**, Onur Alev, Eda Goldenberg and Michael Cheffena, “A Novel Molybdenum Disulfide-Based High-Precision Microwave Sensor for Methanol Gas Detection at Room Temperature,” *IEEE Microwave and Wireless Technology Letters*, vol. 34, no. 6, pp. 691 – 694, 2024 (2024)
- 📄 **Mohammad Mahmudul Hasan**, Michael Cheffena, “Adaptive Antenna Impedance Matching Using Low-Complexity Shallow Learning Model”, *IEEE Access*, vol. 11, pp. 74101 – 74111, 2023 (2023)
- 📄 **Mohammad Mahmudul Hasan**, Michael Cheffena, Slobodan Petrovic, “Physical-layer Security Improvement in MIMO OFDM Systems using Multilevel Chaotic Encryption”, *IEEE Access*, vol. 11, pp. 64468 – 64475, 2023 (2023)
- 📄 Inaamullah Khan, Michael Cheffena, **Mohammad Mahmudul Hasan**, “Data Aided Channel Estimation for MIMO-OFDM Wireless Systems Using Reliable Carriers”, *IEEE Access*, vol. 11, pp. 47836 – 47847, 2023 (2023)
- 📄 **Mohammad Mahmudul Hasan**, Mohammad Mahadi Hasan Foad, “Modified Gamma Correction Companding for PAPR Reduction in OFDM Systems Considering Solid State Power Amplifier and Wireless Channels”, *Circuits, Systems, and Signal Processing*, vol. 37, no. 10, pp. 4431- 4454, 2018 (2018)
- 📄 **Mohammad Mahmudul Hasan**, Mohammad Mahdi Hasan Faisal, “IGCC for PAPR Reduction in OFDM Systems over the Nonlinearity of SSPA and Wireless Fading Channels”, *Circuits, Systems, and Signal Processing*, vol. 35, no. 8, pp. 2855–2880, 2015 (2015)
- 📄 **Mohammad Mahmudul Hasan**, “PAR Reduction in SU/MU-MIMO OFDM Systems using OPF Precoding over the Nonlinearity of SSPA”, *Wireless Personal Communications*, vol. 83, no. 3, pp. 2225-2248, 2015 (2015)
- 📄 **Mohammad Mahmudul Hasan**, “A Novel CVM Precoding Scheme for PAPR Reduction in OFDM Transmissions”, *Wireless Network*, vol. 20, no. 6, pp. 1573-1581, 2014 (2014)
- 📄 **Mohammad Mahmudul Hasan**, “A New PAPR Reduction Scheme for OFDM Systems Based on Gamma Correction”, *Circuits, Systems, and Signal Processing*, vol. 33, no. 5, pp. 1655-1668, 2014 (2014)
- 📄 **Mohammad Mahmudul Hasan**, “A New PAPR Reduction Technique in OFDM Systems Using Linear Predictive Coding”, *Wireless Personal Communications*, vol. 75, no. 1, pp. 707-721, 2014 (2014)
- 📄 **Mohammad Mahmudul Hasan**, “VLM Precoded SLM Technique for PAPR Reduction in OFDM Systems”, *Wireless Personal Communications*, vol. 73, no. 3, pp. 791-801, 2013 (2013)
- 📄 **Mohammad Mahmudul Hasan**, “PAPR Reduction in OFDM Systems Based on Autoregressive Filtering”, *Circuits, Systems, and Signal Processing*, vol. 33, no. 5, pp. 1637-1654, 2013 (2013)

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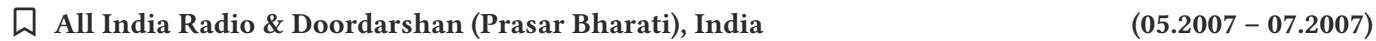
RESEARCH PROJECTS AND INITIATIVES

- 📄 **Machine Learning Assisted Signal Processing for Antenna-Sensor Arrays** (2024 – 2025)
Work Package Member, The Research Council of Norway (Project Number: 353890)
- 📄 **Air Quality Monitoring Exploiting Massive Multiple-Input Multiple-Output Antennas** (2022 – 2025)
Work Package Member, The Research Council of Norway (Project Number: 324061)
- 📄 **University Comprehensive Academic Management- Software Implementation Project** (2020 – 2022)
Project Manager, University of Information Technology and Sciences (UITS)
- 📄 **Higher Education Quality Enhancement Project (HEQEP)** (2017 – 2018)
Member, University Grants Commission, Bangladesh and the World Bank

ACADEMIC LEADERSHIP AND PROFESSIONAL SERVICE

 University of Information Technology and Science, Bangladesh (01.2011 – Present)
✓ <i>Advisor</i> , Information and Communication Technology Cell 01.2018 – Present
✓ <i>Head of the Department</i> , Electronics and Communication Engineering 02.2019 – 04.2022
✓ <i>Director</i> , Information and Communication Technology Cell 12.2020 – 04.2022
✓ <i>Editorial Board Member</i> , Journal of Science and Engineering 10.2020 – 04.2022
✓ <i>Member</i> , Self-Assessment Committee, Internal Quality Assurance Cell (IQAC) 07.2017 – 07.2018
✓ <i>Member</i> , Finance Committee 03.2020 – 04.2022
✓ <i>Member</i> , Academic Council 02.2019 – 04.2022
 Sanyo Engineering & Construction Inc., Japan (04.2018 – 01.2019)
✓ <i>Consultant and Technical Trainer</i> , BJIT Limited, Bangladesh

PROFESSIONAL CERTIFICATIONS

 Brno University of Technology, Czech Republic (08.2024 – 01.2025)
Professional Competence, Electrical Engineering – NV No. 194/2022 Coll.
 GrameenPhone Ltd., Bangladesh (07.2007 – 10.2007)
Intern Engineer, Transmission Planning Division
 All India Radio & Doordarshan (Prasar Bharati), India (05.2007 – 07.2007)
Industrial Training, Broadcasting Corporation of India
 Red Hat Bhubaneswar, India (04.2006 – 07.2006)
Industrial Training, Red Hat Linux RHEL 4

GRANTS & AWARDS

 Research Recognition (06.2025)
Nominated for <i>Best Doctoral Thesis</i> in Sensors Norwegian University of Science and Technology
 Research Grant (08.2024)
Awarded 146 000 NOK The Research Council of Norway
 Chancellor's Gold Medal (12.2010)
Awarded for <i>securing the highest CGPA</i> (10/10) KIIT University, India
 Founder's Gold Medal (12.2010)
Awarded for securing the <i>first position</i> in Master of Technology KIIT University, India
 Chancellor's Merit Scholarship (2004 – 2010)
Awarded for <i>academic excellence</i> , Bachelor's and Master's studies KIIT University, India.

LANGUAGE PROFICIENCY

- ✓ Bengali (native), English (fluent), Hindi (fluent), Arabic (A1), and Norsk (A2/B1)

CITIZENSHIP & RESIDENCY

- 🏠 Bangladeshi (Citizen), Norway (Permanent Residency)

REFERENCES

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